Hysteresis loops of magnetic and nonmagnetic materials

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The hysteresis curves of two different ferromagnetic materials are examined to determine their magnetic properties. For PERMENORM® 5000 H2this results in a remanence of $B_{rem}^{FeNi} = (0.979 \pm 0.022)$ T and a coercivity of $H_{coer}^{FeNi} = (3.137 \pm 0.063)\frac{\mathrm{A}}{\mathrm{m}}$. The magnetic saturation was not reached. For flat steel ST 37 a magnetic saturation of $B_{sat}^{Fe} = (1.4484 \pm 0.0042)$ T, a remanence of $B_{rem}^{Fe} = (0.8544 \pm 0.0059)$ T and a coercivity of $H_{coer}^{Fe} = (190.13 \pm 0.35)\ \frac{\mathrm{A}}{\mathrm{m}}$ are determined. Flat steel has larger hysteresis losses with $\xi_{remag}^{Fe} = (0.203 \pm 0.011)\ \frac{\mathrm{J}}{\mathrm{kg}}$ compared to the losses of Ni-Fe with $\xi_{remag} = (3.69 \pm 0.18)$ J/kg. Finally, the permeability of wood is determined to $\mu_{wood} = (1.5562 \pm 0.0010)$.

Executed: December 8, 2021
First submission: December 15, 2021
Second submission: January 26, 2022

1. Introduction

The most commonly used magnetic materials are ferromagnets. Their behaviour depends not only on the applied magnetic field but also on the history of the material. This phenomenon is called hysteresis and undermines the concept of cause and direct effect. This effect also accounts for the characteristic parameters of ferromagnetics. The remanence is the remaining magnetisation without the field. In order to have no magnetic effect in the material, coercivity is the required outer field strength. These phenomena were tested on different materials. In addition, a wooden core was examined for its magnetic properties.

2. Magnetic model

In this part of the experiment, characteristics of ferromagnetic substances are defined. When a ferromagnetic substance is placed in an alternating magnetic field, certain characteristics can be measured. To visualize these specific characteristics, a model of ferromagnetic needles is placed between two Helmholtz coils. These two coils generate a nearly homogeneous magnetic field between them. Before placing the model inside the magnetic field, the needles are in a fundamental state. In this state the magnetic domains, discovered by Pierre-Ernest Weiss, can be seen. A magnetic domain contains only needles which are oriented in the same direction. This means that the individual magnetic moments are aligned and point in the same direction. A ferromagnetic substance can consist of several magnetic domains that are pointed in different directions. The structure of

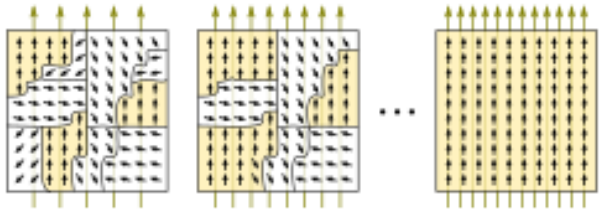

Fig. 1: Shown is the magnetization of a model of magnetic needles. The movements of the magnetic domains are marked. In the end the magnetic saturation is reached [1].

these magnetic domains is responsible for the behavior of a ferromagnet. "Magnetic domains form in materials which have magnetic ordering; that is, their dipoles spontaneously align due to the exchange interaction. These are the ferromagnetic, ferrimagnetic and anti ferromagnetic materials. Para-magnetic and diamagnetic materials, in which the dipoles align in response to an external field but do not spontaneously align, do not have magnetic domains" [1].

Now the applied voltage to the coils is slowly increased, resulting in a stronger field between the coils. One can observe, that the magnetic domains in the model increase in size. The needles begin to slowly rotate to align with the direction of the external field. The borders of the magnetic domains are moving. These borders are called Bloch walls. When all needles point in the same direction as the external field, the ferromagnetic material has reached saturation. The external field can't increase the magnetization of the substance past this saturation, no matter how strong the exter-

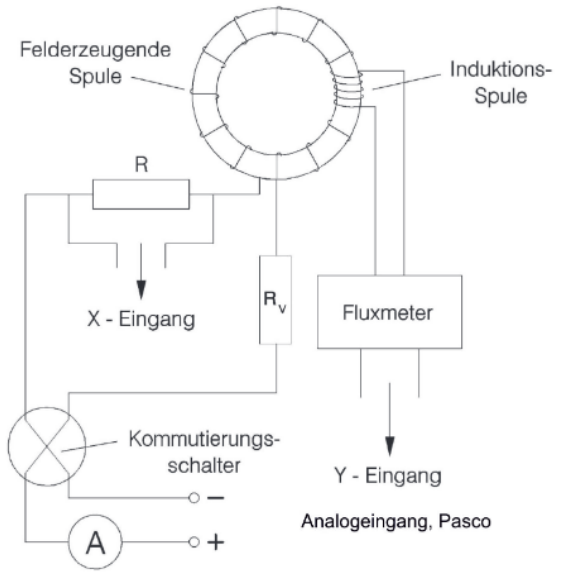

Fig. 2: The construction to measure the hysteresis loops. By varying the current through the field-generating coil, an adjustable magnetic field is generated in the coil core. This connects the induction coil to the former. The magnetic flux density can be measured by integrating the induction voltage. Extracted from [6]

nal field gets. By slowly decreasing the voltage, this process can't be reversed. Some needles may change their direction, but the majority remain aligned in the same direction. Even if there is no longer an external field, the ferromagnetic material is still magnetized. The magnetization left behind when the external field is removed is called the magnetic remanence.

To bring the needles back to the unmagnetized, unorganized fundamental state, the external magnetic field must be rotated. This happens by reversing the polarity. The voltage is slowly increased to neutralize the magnetization of the needles. Applying the voltage generates the coercive field. Thus, the coercivity is the intensity of this field required to demagnetize the material.

By increasing the voltage, the magnetic orientation in the whole model aligns again. Unlike before, the needles now point in the opposite direction. Moreover, by decreasing the voltage, the remanence can be reached again. To demagnetize the model, the voltage is switched off and a permanent magnet is rotated over the needles.

3. Methods of the Experiment

In this experiment, significant characteristics of different materials were computed by measuring hysteresis loops. The construction in figure 2 was used. It shows two ring coils linked by a material. The properties of this material are to be analysed. At the first coil we measure the voltage U_x with the known resistance R. This coil produces the magnetic field strength

$$H = \frac{N_1}{\pi \frac{1}{2}(d_o + d_i)R} U_x, \tag{1}$$

where N_1 is the coil number, d_o is the the outer diameter and d_i is the inner diameter of the first coil. This magnetic field produces a magnetic flux density B through the core of the coils. Changes of the magnetic flux density B induct a voltage surge in the second coil. The flux meter integrates this surge to a value proportional to the B-field. The magnetic flux density is computed as

$$B = \frac{\gamma}{N_2 A} U_y, \tag{2}$$

where γ is the proportionality factor form the flux meter, N_2 is the number of coils and U_y is the measured flux meter voltage. The effective area A is defined as

$$A = \frac{1}{2}(d_o - d_i) \cdot h \cdot f \tag{3}$$

with the height h of the induction coil and the fill factor f of the core inside the induction coil. This computes the effective area influenced by the magnetic field inside the induction coil. Therefore the hysteresis loops can be fully measured by varying the input current and reversing the polarity. An interface (PASCO Universal Interface) shows the hysteresis loops. The recorded data provided by the flux meter is not centered around the zero point. The data is shifted in B-direction with a free constant. To get the desired hysteresis loops the data needs to be centered around the zero point. This is done by placing the zero point in the middle between minimum and maximum recorded voltage. Before doing this shift, it is important to eliminate the time-dependent zero point offset. It can be computed as

$$\Delta U_d = \frac{U_{y,n} - U_{y,1}}{T} \tag{4}$$

where $U_{y,n}$ is the last measured and $U_{y,1}$ is the first measured voltage from the flux meter. The time between these two measure points is T. This zero point offset grows linear with time; therefore each measured voltage can be corrected.

4. PERMENORM® 5000 H2 Fe-Ni-alloy

PERMENORM® 5000 H2 (PH2) is an iron-nickel alloy used industrially. To determine the remanence and the coercivity, a hysteresis loop was measured using a maximum current of about 100 mA. The measurement with this current does not result in saturation. The measured data is shown in figure 3.

The coercivity H_{coer} is given by the intersection with the x-axis, the remanence B_{rem} by the intersection with the y-axis. For determination of the values, regressions

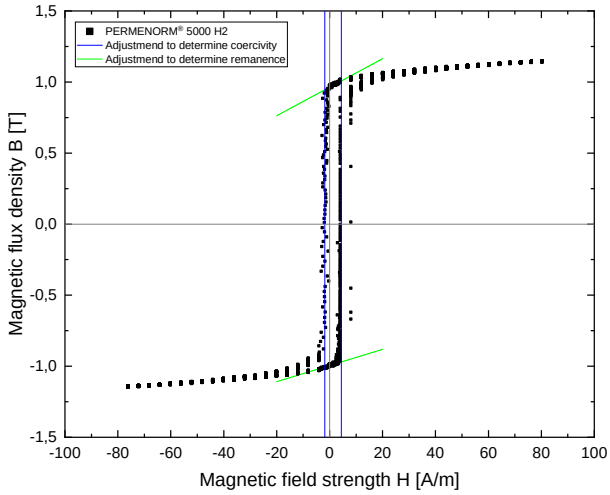

Fig. 3: Hysteresis loop of PERMENORM® 5000 H2 at a maximum current of 100 mA. Regressions were performed to determine remanence and coercivity. The resulting values are $B_{rem}^{FeNi} = (0.979 \pm 0.022)$ T and $H_{coer}^{FeNi} = (3.137 \pm 0.063)\,\frac{\mathrm{A}}{\mathrm{m}}$.

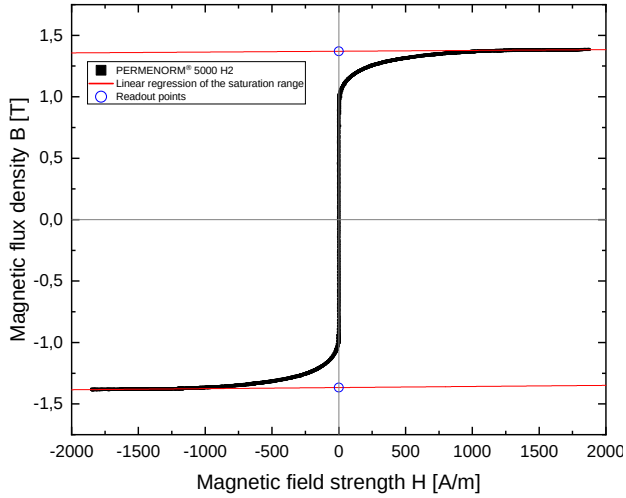

Fig. 4: Hysteresis loop of PERMENORM® 5000 H2 at a maximum current of 2.3A. Regressions were performed to determine the magnetic saturation. The value obtained is $B_{sat}^{FeNi} = (1.369 \pm 0.010)\ T$.

are performed around the intercepts of the x- and y-axis, respectively. The intersection points are determined, and their average value is calculated, taking into account both the errors of the regression and the standard error.

With this method the resulting values are

$$B_{rem}^{FeNi} = (0.979 \pm 0.022)\ \mathrm{T}$$

$$H_{coer}^{FeNi} = (3.137 \pm 0.063)\ \frac{\mathrm{A}}{\mathrm{m}}.$$

In the manufacturer's information, the coercivity of PH2 is given as $H_{lit}^{c} = 5\ \frac{\mathrm{A}}{\mathrm{m}}$ [3] and therefore is not within the margin of error. That stems from the fact that in this measurement saturation was not reached.

Further, the resetting loss $\xi_{remag.}$ can be determined by calculating the enclosed area of the hysteresis loop. The value results to

$$\xi_{remag.}^{FeNi} = (3.69 \pm 0.18) \cdot 10^{-3}\ \frac{\mathrm{J}}{\mathrm{kg}},$$

taking into account an estimated error of 5%.

To reach the saturation range, the measurement was repeated using a current of 2.3 A. The recorded hysteresis curve is shown in fig. 4. By performing a regression of the linear, almost constant parts of the curve, the magnetic saturation can be read out as the y-axis intersection of the regression. Averaged over the two reading points, the resulting value is

$$B_{sat}^{FeNi} = (1.369 \pm 0.010)\ T.$$

The manufacturer's specified value for the magnetic saturation is $B_{sat}^{lit} = 1.55$ T. This does not match the calculated value within its error. The reason might be that the saturation range has not yet been reached, even though the data suggest so. The slope of the regression expresses the quality in which the saturation range was reached. The connection between the permeability and the B- and H-field is given as

$$\mu = \mu_0 \mu_r = \frac{B}{H}. \tag{5}$$

Where $\mu_0 = 1.25663706212 \cdot 10^{-6}\ \frac{\mathrm{N}}{\mathrm{A}^2}$ [4] is the magnetic constant and μ_r the permeability. When saturation is reached μ_r converges to 1 and therefore the expected value would be equal to μ_0. The weighted average of both slopes of the regressions is $\mu_{NiFe} = (7.18 \pm 2.23) \cdot 10^{-6}\ \frac{\mathrm{N}}{\mathrm{A}^2}$, and is therefore significantly higher than μ_0. This is an indicator that saturation has indeed not yet been fully reached. This shows that both coercivity and saturation induction were estimated to be too small.

5. Flat steel

As in the previous measurements, a complete hysteresis curve was recorded for the flat steel core seen in figure 5. For reference, the magnetic saturation, the remanence, the coercivity and the resetting loss was determined to be

$$B_{sat}^{Fe} = (1.4484 \pm 0.0042)\ \mathrm{T}$$

$$B_{rem}^{Fe} = (0.8544 \pm 0.0059)\ \mathrm{T}$$

$$H_{coer}^{Fe} = (190.13 \pm 0.35)\ \frac{\mathrm{A}}{\mathrm{m}}$$

$$\xi_{remag}^{Fe} = (0.203 \pm 0.011)\ \frac{\mathrm{J}}{\mathrm{kg}}.$$

The errors are the result of the maximum error estimations of the input data and the regressions.

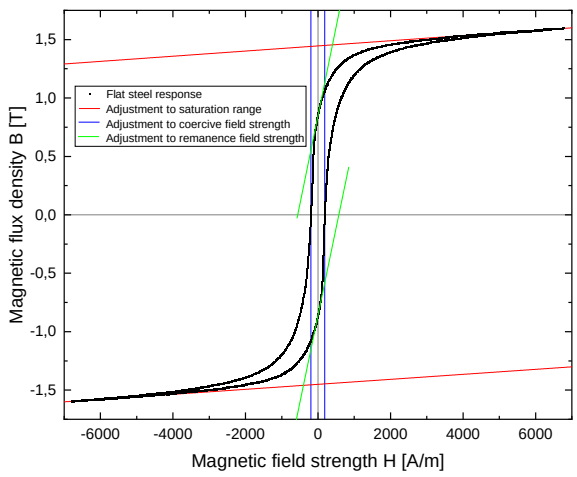

Fig. 5: The hysteresis curve of the flat steel is shown. Regressions were performed to determine remanence, coercivity and magnetic saturation. The resulting values are $B_{rem}^{Fe} = (0.8544 \pm 0.0059)$ T, $H_{coer}^{Fe} = (190.13 \pm 0.35)\ \frac{\mathrm{A}}{\mathrm{m}}$ and $B_{sat}^{Fe} = (1.4484 \pm 0.0042)\ \mathrm{T}\frac{\mathrm{A}}{\mathrm{m}}$.

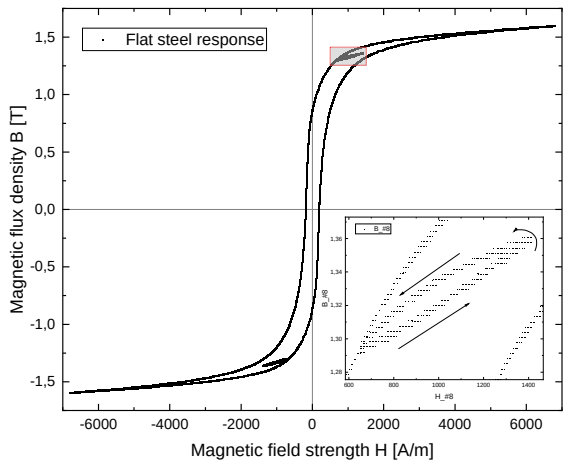

Fig. 6: Shown are two loops that are created by briefly reversing the direction of the field. The symmetry was caused by the choice of reversal points. In the inlet you can see the enlargement of the upper loop. The direction of the loop is counterclockwise.

If the direction of the hysteresis curve changes briefly, internal hysteresis loops can be observed. As can be seen in figure 6, the hysteresis curve was recorded completely and at two points the direction was inverted and the field strength slightly increased before continuing with the the curve. In the inlet, it can be seen in more detail, that the loops have been passed through in a counterclockwise direction. The apparent symmetry of the loops comes stems the choice of turning points and is not an effect of the material.

This effect can be utilised to set a certain value as remanence by rotation around it. In particular the remanence of zero is of interest. This process is called demagnetisation. An attempt to demagnetise the flat steel core can be seen in figure 7.

Looking at the spiral-shaped curve, especially of the inlet, it can be seen that the demagnetisation has not succeeded well. A better result would have been achieved if the step size of the maximum field strength had been smaller. In addition, the penultimate reversal point should have been chosen after the coercivity had been reached. Therefore a residual remanence remains, which is considerably lower than the remanence from magnetisation saturation.

6. Permeability of wood

In this part we computed the permeability of a non-magnetic substance. Therefore, the coil with a wooden core was used. The wood core filled the whole space inside the coil. The specifications of the coil can be seen in table 1. To get the permeability of the filling substance, the flux meter measured the magnetic flux density $\vec{B}$,which is dependent on the magnetic field strength $\vec{H}$ according to

$$\vec{B} = \mu_0(\vec{H} + \vec{M}) = \mu_0(1+\chi)\vec{H} = \mu_0\mu_r\vec{H} \qquad (6)$$

where μ_0 is the permeability constant, χ is the magnetic susceptibility and $\vec{M}$ is the magnetization of a material. The measurements of the flux meter were transformed according to equations 2 and 4. The errors of the used measuring devices are stated in appendix B. The figure 8 shows the magnetic flux density over the magnetic field strength. No hysteresis loop can be seen, as the filling substance of the coil is not a ferromagnetic substance. The gradient of figure 8 equals the permeability μ according to equation 5 and is

$$\mu = (1.9553 \pm 0.0013) \cdot 10^{-6}\ \frac{N}{A^2}.$$

The permeability constant is $\mu_0 = 1.25663706212 \cdot 10^{-6}\ \frac{N}{A^2}$ [4]. Therefore the permeability of wood is

$$\mu_r = \mu_{wood} = (1.5562 \pm 0.0010).$$

This does not align with the expected value of $\mu_{wood,lit.} = 0.9999994$ [5]. Wood is a diamagnet. The differences can be explained by several factors. The construction of the experiment can lead to differences in the data collected. Moreover, the permeability of wood is never differed by the kind of wood used for the experiment. In conclusion this shows that the permeability of wood is smaller than these of the examined ferromagnetic substances.

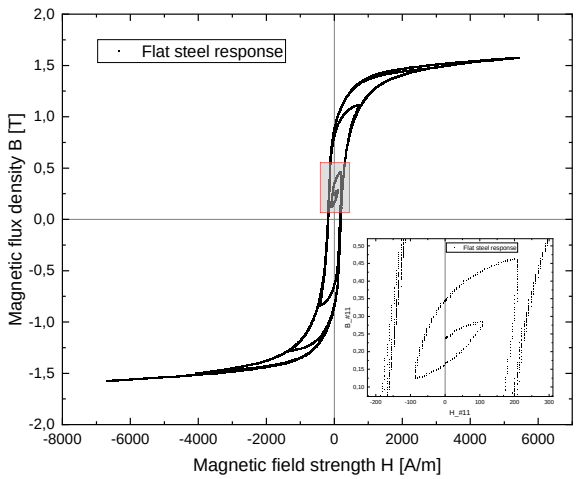

Fig. 7: Demagnetisation of the iron core from saturation by repeatedly reversing the polarity of the magnetic field with decreasing magnitude. Afterwards the magnetic field rises again to saturation magnetisation. The central area is shown enlarged in the inlet. It can be seen, that the ferromagnet was not completely demagnetised.

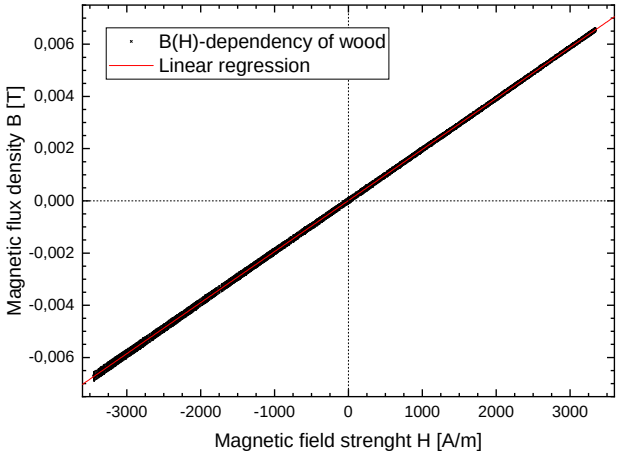

Fig. 8: The measured magnetic flux density created by the induction of a coil with a wooden core. No hysteresis loop can be detected and a linear regression was added. The permeability of $\mu = (1.9553 \pm 0.0013) \cdot 10^{-6}\ \frac{N}{A^2}$ can be selected.

7. Summary

The hysteresis loops for two ferromagnetic substances and for wood as a nonmagnetic substance, for reference were measured. From the graphic evaluation of the hysteresis loops, several characteristic values of the substance could be seen. For the flat steel (ST37K) the characteristic values are the magnetic saturation $B_{sat}^{Fe} = (1.4484 \pm 0.0012)$ T, the remanence $B_{rem}^{Fe} = (0.85436 \pm 0.00041)$ T, the coercivity $H_{coer}^{Fe} = (190.13 \pm 0.35)$ A/m and the hysteresis loss $\xi_{remag} = (0.2033 \pm 0.0029)$ J/kg. For the Ni-Fe-alloy, PERMENORM® 5000 H2 the characteristic values are the remanence $B_{rem}^{FeNi} = (0.979 \pm 0.022)$ T, the coercivity $H_{coer}^{FeNi} = (3.137 \pm 0.063)\frac{\mathrm{A}}{\mathrm{m}}$ and the hysteresis loss $\xi_{remag.}^{FeNi} = (3.69 \pm 0.18) \cdot 10^{-3}\ \frac{\mathrm{J}}{\mathrm{kg}}$. The permeability of wood is $\mu_{wood} = (1.5562 \pm 0.0010)$. All measured hysteresis loops as the B(H)-dependency of wood follow the qualitative expected trend. Furthermore, the irreversible effect of a change of direction in a hysteresis loop was shown with the flat steel (ST37K). This effect can be used for demagnetisation.

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A. Supplementary

In this experiment, magnetic fields will be measured. A flux meter will be recording the experimental data. To understand the function of this machine a short introduction will be given now. At first, the function of an ideal flux meter will be discussed. To measure the magnetic fields the effect of induction will be used. A magnetic field produces a proportional induction effect in an induction coil. The circuit diagram of a flux meter with an ideal amplifier is shown in figure 9. According to the Kirchhoff´s circuit laws the following equations can be made;

$$I_e = \frac{U_e}{R} = -I_c \tag{7}$$

$$U_c = U_a \tag{8}$$

$$I_c = C\ \frac{d\mathrm{U}_c}{dt} = C\ \frac{d\mathrm{U}_a}{dt} \tag{9}$$

With these equations we can compute that

$$U_a = \frac{-1}{RC} \int U_e\ dt. \tag{10}$$

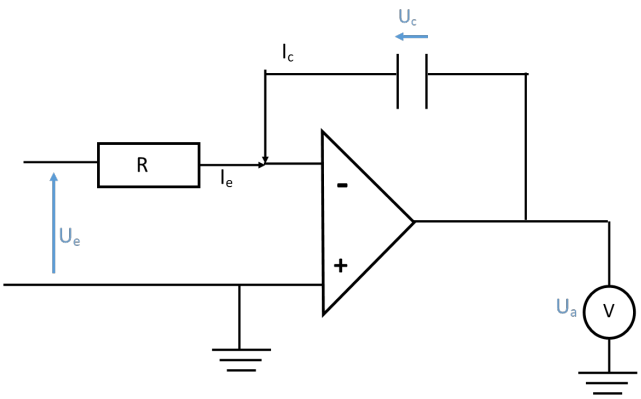

Fig. 9: This figure shows the schematic buildup of a flux meter. Orientated on [2].

With the output voltage U_a the flux meter can compute the input voltage impact $\int U_e \, dt$. This is just an idealised machine. A real flux meter has a few interfering factors. The biggest problem with the flux meter is the offset voltage U_{off} and the offset electricity I_{off} of the amplifier inside the flux meter. Considering these parameters we get the realistic equation of

$$U_a = -\int \left(\frac{U_e + U_{off}}{RC} + \frac{Ioff}{C}\right) dt. \tag{11}$$

These disruptive factors can easily be noticed while measuring hysteresis curves of nonmagnetic material. If a measure is taken over a long time, the branches of the hysteresis curve are not closed in the stating or end point. The reason for this is capacitor inside the flux meter (see figure 9). Nonmagnetic material doesn't have a remanence, therefore this effect can be seen very clearly in the graphical output of the flux meter.

B. Error discussion

There was no inaccuracy for the flux meter given by the developer. Therefore, the inaccuracy was assumed to be 1%. There was an inaccuracy given for the interface. According to the developer it is $\sigma_U = 1.22$ mV. For all figures, the error bar can not be seen, therefore the "read-off-error" was significantly higher and taken as inaccuracy of the stated values in this paper.

C. Coil data

	Fe-Ni	flat steel	wood
d_o	260 mm	260 mm	260 mm
d_i	220 mm	220 mm	220 mm
N_1	605	1010	1010
N_2	80	53	250
h	30.0 mm	30.0 mm	30.0 mm
f	90%	100 %	100 %
R	0.20 Ω	0.20 Ω	0.20 Ω

Table 1: Values of the coils used in this experiment.